AI-Based SEO, AEO and GEO Readiness Audit System

Project Overview for Supervisor Review

English | October 2026 | Covers: SEO module, AEO Answerability Analyzer, GEO module, Entity Graph

About this document: an overview of the proposed project: what the system does, how it works, how it will be evaluated, and the planned scope. It covers the SEO module, the AEO Answerability Analyzer, the GEO module and the Entity Graph.

1. The project at a glance

In one sentence: our system crawls a website and tells the owner how ready it is for Google and AI search, and what to fix first.

Problem: small business owners do not know what is wrong with their website. Existing tools (Screaming Frog, Ahrefs) are expensive or technical, and mostly check traditional SEO only.

Our solution: a report in plain language, three separate readiness scores (SEO, AEO, GEO), the reason behind every issue, and AI-written, site-specific recommendations.

Our system DOES	Our system does NOT
Crawl a website and audit it	Fix the website itself (the owner does that)
Give three readiness scores	Guarantee Google rankings or ChatGPT citations
Give recommendations and schema/FAQ drafts	Invent business facts
Show before/after improvement on re-audit	Give official Google scores (our scores are our own assessment)

Think of the system as a **doctor**: it does the check-up and writes the prescription (recommendations). Taking the medicine (applying the fixes) is the patient's (website owner's) job.

2. Key terms (glossary)

Term	Meaning
SEO	Search Engine Optimization: getting a site to appear in Google results.
AEO	Answer Engine Optimization: structuring content so search and AI engines can pull a direct answer from it.
GEO	Generative Engine Optimization: making content easy for generative AI (ChatGPT, Gemini) to understand, use and cite in its answers.
LLMO	Large Language Model Optimization: helping AI models represent a brand correctly. Overlaps heavily with GEO (section 7).
Cite	To name a source. An AI answer showing a site's name or link.
NLP	Natural Language Processing: teaching computers to understand human language.
LLM	Large Language Model (such as Claude or ChatGPT): reads and writes text.
Entity	A named thing: a business, city, product or person.
NER	Named Entity Recognition: finding named things in text and labelling them.
Meta description	The 120-160 character summary shown under the title in search results.
ALT text	A text description of an image; Google and screen readers understand images through it.

Term	Meaning
Index / noindex	Index: Google saved the page in its database. A noindex tag tells Google not to save it.
Schema (JSON-LD)	Structured data code that tells engines what is on a page (business, FAQ, product).
robots.txt / sitemap.xml	robots.txt: rules for crawlers. sitemap.xml: a list of the site's pages.
Canonical	A tag saying which page is the original among duplicates.
Ground truth	The verified "correct answer" we compare our system's output against.

3. SEO, AEO, GEO: three different questions

	Question it answers	Example checks
SEO	Can Google find and rank my site?	Title, meta, H1, ALT, speed, broken links, sitemap
AEO	Does my site answer questions clearly?	FAQ, question headings, answerability test
GEO	Can AI tools understand and use my site?	AI bots allowed, schema, entities, brand consistency

The most important distinction: SEO, AEO and GEO are **optimization techniques** (we apply them to the website). Google results, snippets, ChatGPT answers and citations are the **output of the engines** (the engines produce them, not us). Our app only **audits**.

SEO is the foundation; AEO and GEO sit on top of it. If SEO is weak, AEO and GEO help less. According to Google's guidance (cited in the FYP ideas PDF), there is no special optimization for its AI features and basic SEO still matters most. AEO, GEO and LLMO have no official ranking-factor list, so call our checks "best-practice signals", not "ranking factors".

4. How the system works (full flow)

Step	Task	Detail
1	URL input	User enters a URL. The system checks the format, reachability and HTTPS.
2	Crawling	Start at the homepage, check robots.txt and the sitemap, queue internal links. Page limit (20-30), timeout and request delay.
3	Data extraction	From each page: title, meta, headings, images/ALT, links, canonical, viewport, robots tag, schema and body text.
4	SEO rules	Fixed checks. Every failed check becomes an issue (Critical/Warning/Info).
5	AEO module	Rules plus the Answerability Analyzer (section 6).
6	GEO module	AI bots, schema, trust signals, Entity Graph (sections 7-8).
7	Scoring	Separate SEO, AEO and GEO scores plus an overall score.
8	LLM recommendations	Site-specific fixes from the findings (new title, FAQ draft, schema draft).
9	Report	Dashboard, issues list, graph, PDF export, history and before/after.

5. SEO module (the foundation)

Rule-based analysis means: the crawler extracts the page data, then each check is a small function that returns an issue, or None if everything is fine.

Category	Checks
On-page	Title missing/empty/length, meta description missing/empty/length, H1 missing or multiple, heading order
Images	ALT missing, oversized images
Links	Broken links (404), empty anchor text, crawlable links
Technical	HTTP status, noindex, canonical, robots.txt, sitemap.xml, HTTPS, viewport, schema
Performance	Page load time, Core Web Vitals (PageSpeed API)

Severity	Meaning	Examples
Critical	Directly harms indexing or ranking	Title missing, noindex set, page returns 404, no HTTPS
Warning	Important, but the page can still be indexed	ALT missing, thin content, broken link, no sitemap
Info	Minor improvement	Slightly long meta description, add schema

Rules learned from real testing (books.toscrape.com)

- **Check the status code first.** A 404 error page still has a title and an H1. Without a status check, the system would treat a broken page as healthy. 200 -> run the rules; 404/410 -> report "broken page"; 301/302 -> follow the redirect.
- **"Missing" and "empty" are different.** The site had <meta name="description" content="" />: the tag exists but the text is blank. A rule that only asks "is the tag there?" would wrongly pass this page.
- **Use strip() on the title and other text.** The title in the source had new lines and spaces around it; calling len() directly gives the wrong length.
- **Lowercase the robots tag before comparing.** The site had NOARCHIVE,NOCACHE. That is not noindex; only treat noindex as "blocked", otherwise you create false alarms.
- **A 404 on robots.txt means "no restrictions".** It is not a serious error, just a small issue. The GEO check "AI bots blocked" should pass in that case.

```
def check_meta_description(page):
    tag = page.get("meta_description")      # None = tag missing
    if tag is None:   return issue("meta_description_missing", "warning")
    if tag.strip()=="": return issue("meta_description_empty", "warning")
    n = len(tag.strip())
    if n < 70:  return issue("meta_description_short", "info")
    if n > 160: return issue("meta_description_long", "info")
    return None
```

6. AEO module (new: Answerability Analyzer)

Goal: check whether the page contains clear, direct answers to questions, so answer engines can extract them.

Part A: AEO rules

- Is there FAQ content or FAQ schema?
- Are headings written as questions? ("Which cakes can I order online?")
- Does a short, direct answer (about 40-60 words) follow each question?
- Are lists and tables used instead of long paragraphs?

Part B: Answerability Analyzer (the real AI work)

Step	What happens
1. Questions	An LLM writes questions from the page text (only from the page text).

Step	What happens
2. Answer test	For each question, check whether the page contains a clear answer (retrieval + extraction).
3. Score	Clarity and completeness: Clear / Partial / No answer, plus an overall answerability score.
4. Suggestions	If there is no answer, give a rewrite suggestion **without inventing facts**. Anything not on the page becomes a placeholder.

Example (Ali Bakery): 6 questions were generated. 2 had a clear answer (which cakes are sold, where delivery goes), 1 was partial (same-day delivery but no timing), 3 had no answer (lead time, opening hours, delivery charge). Suggestion: "Q: How long does a custom cake order take? A: [your confirmed lead time]". The brackets mean the owner fills in the real fact.

Important: keep the LLM grounded in the page text only. If it makes up an address or opening hours, that is a false fact, and that would be a serious flaw in the system.

7. GEO module (and LLMO)

Goal: check whether generative AI (ChatGPT, Gemini, Perplexity, Google AI features) can understand our site and use or cite it.

Check	Severity	Detail
AI bots blocked in robots.txt	Warning	"Disallow: /" for GPTBot, PerplexityBot, ClaudeBot. If there is no robots.txt, bots are not blocked (check passes).
Organization/LocalBusiness schema	Warning	Whether the business details are machine-readable.
Author, about, contact info	Warning	Trust signals.
Specific facts vs vague claims	Info	"Best cakes in town" has no proof; "5,000+ orders since 2015" does.
Last-updated date	Info	Freshness signal.
Brand consistency	Warning	Are name, address and phone the same on every page? (via the Entity Graph, section 8)

How LLMO relates to GEO

LLMO means helping AI represent the brand correctly. GEO means getting the site used or cited in AI answers. Their goals and checks overlap heavily (clear facts, schema, entities, consistent information), and neither has one standard definition. So we **did not make LLMO a separate pillar**; it is part of the GEO module.

8. Entity Graph (new part of GEO)

An **entity** is a named thing (business, city, product). A **graph** shows the relationships between them. Goal: "If an AI reads only this website's text, what will it learn about the business, and what will it miss?"

```
Ali Bakery --sells--------> Birthday Cakes
     |--located in-----> Islamabad
     |--phone----------> 0300-1234567
     |--delivers to----> F-10, F-11
     |--address--------> ?  (missing)
```

Step	Task	How
1	Extract entities	Phone/email: regex. City: spaCy. Business name: title, schema or LLM. Services: LLM.
2	Normalize names	Fuzzy matching (rapidfuzz). "Ali Bakery" vs "Ali Bakers" count as one entity, but two spellings on the site is an issue.
3	Relationships	Fixed, predefined relations (located_in, sells, phone, delivers_to). Full relation extraction is future work.

Step	Task	How
4	Graph	NetworkX (no separate database or Neo4j needed).
5	Missing entities	Match against an expected list: name, address, phone, email, hours, services, area served.
6	Schema draft	LocalBusiness JSON-LD with **only facts found on the site**. Missing items are listed separately for the owner.
7	Report	Small graph picture, missing list, name warning, schema draft.

NER limitation: spaCy recognizes famous places (Islamabad) well, but it sometimes mislabels or misses small local business names ("Ali Bakery") and Roman Urdu names. So we do not rely on spaCy alone; regex, title/schema and an LLM are used together. This limitation should be reported honestly in the evaluation.

9. Scoring

Pillar	Proposed weight	Built from
SEO	70%	On-page 20, Content 15, Technical 15, Performance 10, Images+Links 10
AEO	15%	AEO rules + answerability score
GEO	15%	AI access, schema, trust signals, entity completeness

- Every issue has a severity; a category score is the percentage of passed checks, weighted by severity.
- Grades: 0-49 Poor, 50-74 Needs Work, 75-89 Good, 90+ Excellent.
- **The score always comes from rules and measured checks**, never from the LLM (so it is explainable and repeatable).
- The weights are proposed and can be adjusted on the supervisor's advice.

The competitor problem: on one site, three apps gave 97, 92 and 48, and none explained how the score was built. Our system shows the reason for each score (severity and weight).

10. The role of AI: rules, NLP, LLM

Layer	What it does	Example
Rules	Fixed checks, scoring	Is the meta description missing?
NLP	Analyzes text	Keywords, readability, duplicate content, entity extraction, title-vs-content match
LLM	Generates new text	Writing questions, recommendations, FAQ and schema drafts

Controlling the LLM

- Prompts grounded in crawled data only, with JSON output.
- No invented facts: use a placeholder where a fact was not found.
- Keep the API key in the backend only. Measure cost per audit.
- If the LLM is down, still show rule-based recommendations.

11. Technologies, database and folders

Layer	Tools
Frontend	React
Backend	Python + FastAPI (crawl as a background task, status polling)
Crawler	requests + BeautifulSoup (or Scrapy); Playwright later (JavaScript sites)

Layer	Tools
Database	PostgreSQL
NLP	scikit-learn (TF-IDF), textstat, spaCy, rapidfuzz, sentence embeddings
Graph	NetworkX
LLM	Existing LLM API
Extra	Google PageSpeed Insights API, Docker

Database tables: users, audits, pages, issues, recommendations, scores, **entities** (audit_id, type, value, source_page), **questions** (page_id, question, answer_found, clarity_score).

```
seo-audit/
  backend/app/
    api/  crawler/  rules/ (on_page, images, technical, aeo, geo)
    aeo/ (question_gen.py, answer_test.py)
    entities/ (extract.py, normalize.py, graph.py, schema_draft.py)
    nlp/  ai/  scoring/  models/  report/
  frontend/src/pages/ (NewAudit, Report, AEO, EntityGraph, History)
  docker-compose.yml
```

12. Functionalities

Side	Features
User	Register/login; start an audit from a URL; live progress; report (overall + SEO/AEO/GEO scores); issue filters (severity/pillar); Quick Wins; AI recommendations; AEO answerability tab; Entity graph tab; PDF download; audit history and before/after comparison
Crawler	URL validation, robots.txt/sitemap parsing, link discovery, duplicate removal, page limit, redirect chain
Admin (optional)	Edit rules and weights, manage users/audits, monitor API usage

13. How we will check accuracy (Evaluation)

Do not compare score with score. The real proof is whether the issues our system reports were **actually correct**.

Step	Task
1	Choose 10-15 websites (good, weak, different platforms, 1-2 you can fix yourselves). Respect permission.
2	Use only the checks that other tools also perform.
3	Build ground truth: Lighthouse/PageSpeed + browser (Ctrl+U) + (optional) Screaming Frog. If they disagree, the manual check is final.
4	Run our system and export its issues.
5	Compare issue by issue: TP (correctly found), FP (false alarm), FN (missed).
6	Precision = TP/(TP+FP); Recall = TP/(TP+FN).
7	AEO/GEO: two reviewers with a manual rubric; measure their agreement. Report results as "according to our rubric".
8	Have 3 people rate AI suggestions 1-5 (relevant, accurate, easy to apply) and count invented facts.
9	Entity extraction: write the entities of 5-10 pages by hand, then match against the system (precision/recall).
10	Before/after: apply fixes on a practice site and re-audit.

Worked example: books.toscrape.com (practice site)

Check	Verified result	What the system should do
Title	Present, 40 chars	No issue
Meta description	Empty (content="")	Issue
H1	1 ("All products")	No issue
Canonical	Missing	Issue (Info)
Viewport	Present	No issue
Schema	None (Rich Results Test + source)	Issue
robots.txt	Missing (404)	Issue
sitemap.xml	Missing (404)	Issue
noindex	Not set (NOARCHIVE,NOCACHE present)	No issue
Image ALT, HTTP status	Pass (Lighthouse)	No issue

Lighthouse desktop SEO score was 91, with one failed audit: meta description. **Expected result:** the system reports 5 issues and stays quiet on the other 6. This is one site's row set; the same process repeats on the other sites.

14. Competitor comparison (3 mobile apps)

This comparison is based only on the screenshots and recordings we reviewed; we did not see every tab of every app, so "not seen" does not mean "does not exist".

Feature	Seosaner	SEO Meta Tags Analyzer & Audit	SEO Audit	Our system
Score (same site)	97	92	48	Explainable
Compare URLs / Bulk / Sitemap crawler	Not seen	Yes	Full-site audit yes	Nice-to-have
History	Not seen	List only	Trend (after 2 checks)	Before/after compare
Export	CSV	Text share	Not seen	PDF
GEO / AI	Not seen	Not seen	"SEO / GEO", Agent SEO	Yes
AEO	Not seen	Not seen	Not seen	Yes (answerability)
Separate SEO/AEO/GEO scores	No	No	Not seen	Yes

Observation: among the 3 apps reviewed, we did not find AEO readiness combined with three separate, explainable scores.

15. Scope and work split

Tier	What
Must	Crawler, SEO rules, SEO score, report
Should	AEO rules + answerability test, basic GEO checks, LLM recommendations, PDF
Nice	Entity graph (small version), history compare, AI Visibility Test
Skip (for now)	Neo4j, competitor entity comparison, JS rendering (Playwright), every AI engine integration, bulk URLs

Member	Proposed work
Member 1	Crawler + SEO rule engine
Member 2	AEO analyzer, entity extraction and graph, LLM integration
Member 3	Frontend, FastAPI, database, PDF report